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COMMERCIAL ITEM DESCRIPTION
STAIRCASE, AIRCRAFT BOARDING, STANDARD TRUCK

The General Services Administration has authorized the use of this commercial item description for all federal agencies.

1. SCOPE.

1.1 Scope. This commercial item description (CID) covers the general requirements for passenger-boarding, staircase trucks servicing military and commercial aircraft on Government installations.

2. SALIENT CHARACTERISTICS.

2.1 Aircraft staircase description. The commercial, chassis-mounted vehicle shall be capable of servicing multiple aircraft classifications to safely load and unload passengers. It shall have the ability to quickly pull up and back away from aircraft. The lift system shall operate smoothly.

2.1.1 Modified commercial product. The aircraft staircase, at a minimum, shall be the manufacturer's standard commercial product modified as necessary to be in accordance with the requirements specified herein. Additional or complimentary features which are not specifically prohibited by these specifications, but which are part of the manufacturer's standard product, shall be included in the unit being furnished.

2.1.2 Aircraft compatibility. The aircraft staircase shall be compatible with the following aircraft and shall ensure that the nose of the chassis does not come within 12 inches of the aircraft.

Aircraft Accommodated	Height Range (inches)
B-707, B-727, B-737, B-747, B-757, B-767, B-777, A-300, A-320, A-340, and their military derivatives.	96 - 228

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2.2 Design and construction. The aircraft staircase shall be designed and constructed in accordance with SAE AIR6133 and SAE ARP1247 and shall promote cost-effective, life-cycle sustainability by addressing considerations, such as incorporating open standards while satisfying system performance requirements. The aircraft staircase shall be designed for safe operation with the rated-capacity load at all lift heights and positions that the truck is capable of achieving. The aircraft staircase shall be constructed so that parts do not work loose while in service. It shall withstand the loads, shocks, strains, vibrations, and other conditions incident to operation, shipping, and storage without permanent distortion and with minimum loss of time for maintenance, repair, and servicing. It shall be weatherproof and designed to prevent the intrusion of water and sand into critical operating components. The aircraft staircase design shall ensure safe, efficient, and economical operation. All assemblies, controls, and installed equipment shall be located so that there is no adverse interference with each other or with the operation, and shall be readily accessible for maintenance, operation, removal, and replacement.

2.2.1 Materials, protective coatings, and finish.

2.2.1.1 Protective coatings. Materials that deteriorate when exposed to sunlight, weather, or operational conditions normally encountered during service shall not be used, or shall have a means of protection against such deterioration that does not prevent compliance with the performance requirements specified herein. Protective coatings that chip, crack, or scale with age or extreme climate conditions, or when exposed to heat shall not be used. Ferrous structures and surfaces shall be primed with corrosion-inhibiting primer in accordance with MIL-DTL-53022, type II, or a commercial equivalent. Raw metal edges shall be coated with the primer before topcoating. Topcoat shall be polyurethane. Fasteners, handles, and fittings used in the assembly of the aircraft staircase shall also be primed and painted. Surface preparation and pretreatment shall be in accordance with the respective primer and topcoat specifications. Structures shall be cleaned, degreased, and blasted prior to priming; primer shall be applied before any oxidation or rusting occurs. Topcoat shall be applied to a dry-film thickness of 1.6 to 2.4 millimeters (mm) in all instances, regardless of the primer system utilized. The coating shall be free from runs, sags, orange peel, or other defects.

2.2.1.2 Dissimilar metals. Dissimilar metals, as defined in MIL-STD-889, shall not be in contact with each other. Metal plating or metal spraying of dissimilar base metals to provide electromotive-compatible abutting surfaces is acceptable. The use of dissimilar metals is permitted only when separated by suitable insulating material, except in systems where bridging of insulation materials by an electrically-conductive fluid can occur. Sealants or gel-type gasket materials shall be used between faying surfaces and butt joints.

2.2.1.3 Finish. The exterior finish color of the aircraft staircase shall be a standard, commercial white. Exterior surfaces shall be prepared, primed, and painted with polyurethane paint. The interior of all compartments shall be painted with an impact-resistant, textured coating that resists stains, scuffs, chips, and scratches. All exterior surfaces, excluding all normally bright-metal and anodized parts, shall be painted body color. The chassis frame and running gear may be black. All interior surfaces visible with any compartment door open, excluding the interior of the cab, shall also be painted body color. This includes compartment shelves and mounting hardware, but does not include items mounted in the compartments.

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2.2.1.4 Rustproofing. The vehicle chassis and cab (at minimum: seams, welds, locations where dissimilar metals contact, etc.) shall be rustproofed and undercoated in accordance with FED-STD-297, or a commercial equivalent. When specified, the level of corrosion prevention shall be upgraded to tropical (see 6.2). Tropical protection shall be corrosion-protected with a thick, hard, black, commercially-available, corrosion-preventive compound (CPC) that is difficult to remove. Application shall include the exposed chassis-frame metal underneath the aircraft staircase (excluding the drive train), the inside of the wheel wells, and behind the fender skirts. In addition, aircraft staircases that are to be shipped overseas shall be temporarily corrosion-protected from the effect of salt spray, both on the ship and at port, with the application of a thin, soft-film, semi-transparent, commercially-available CPC that is easily removable by washing or with low-pressure steam. Application of temporary CPC shall be to the exterior of the aircraft staircase, in critical or corrosion-sensitive areas of the vehicle. The Air Force vehicle maintainers overseas shall remove this temporary CPC on receipt from the shipper. The technical manual shall reflect the temporary CPC removal process.

2.2.1.5 Exclusion of water. The design of the vehicle shall be such as to prevent water leaking into, or being driven into, any part of the vehicle interior when either in an operating or travelling configuration. All windows, doors, panels, covers, etc., shall be provided with sealing arrangements such that the entry of water is minimized when these items are correctly closed. Particular care shall be taken to prevent wetting of equipment and heat and sound proofing materials. Sharp corners and recesses shall be avoided so that moisture and solid matter cannot accumulate to initiate localized attack. Sealed floors with suitable drainage shall be provided for storage compartments, engine compartments, and other areas in the vehicle that could collect and retain water.

2.2.1.5.1 Fluid traps and faying surfaces. There shall be no fluid traps on the aircraft staircase. Faying surfaces of all structural joints, except welded joints, shall be sealed to preclude fluid intrusion.

2.2.1.5.2 Drainage. Drain holes shall be provided to prevent collection or entrapment of water or other unwanted fluid in areas where exclusion is impractical. All designs shall include considerations for the prevention of water or fluid entrapment, and ensure that drain holes are located to effect maximum drainage of accumulated fluids. The number and location of drain holes shall be sufficient to permit drainage of all fluids when the unit is in a 1-degree incline in any plane. The minimum size of the drain holes shall be 0.25 inch.

2.2.1.5.3 Ventilation. Ventilation shall be sufficient to prevent moisture retention and buildup.

2.2.2 Markings. All external devices which require an operational or maintenance interface shall be marked in accordance with MIL-STD-130. Markings shall be 1-inch high block letters, unless prohibited by the available space. In such cases, the markings shall be the largest size possible. Information and Caution markings shall be Lusterless Black, color number 37038 of SAE AMS-STD-595, and Warning and Danger markings shall be Lusterless Red, color number 31136 of SAE AMS-STD-595. Plates shall not fade due to climate exposure. The center of gravity shall be

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stenciled on the unit within 1 inch of the calculated center of gravity. Tire inflation pressure shall be marked as near to each tire as practicable.

2.2.3 Identification and information plates.

2.2.3.1 Identification plate. An identification plate, in accordance with MIL-STD-130, shall be securely attached to the aircraft staircase in a readily-accessible location. The identification plate shall contain the following information: manufacturer's name, model number, serial number, contract number, registration number, date of delivery, warranty expiration date or mileage, platform and step load limits, gross vehicle weight (pounds), fuel type, speed-limit restrictions, Commercial and Government Entity (CAGE) code, and National Stock Number (NSN). The aircraft staircase and any of its components, for which the Government's unit cost is more than $5,000 or the procuring agency determines is mission essential, is serially managed by the Government and shall have Unique Identification (UID) (also known as Item Unique Identification (IUID)) information permanently affixed on or near the respective identification plate(s), marked in accordance with MIL-STD-130. The UID information shall be included as both a bar code and legible markings. Plates shall not fade due to climate exposure.

2.2.3.2 Transportation data plate. A transportation data plate shall be securely attached to the aircraft staircase in a readily-accessible location. The plate shall contain at least the following information:

a. Side and rear silhouette views of the aircraft staircase.

b. Horizontal and vertical location of the center of gravity of the aircraft staircase in air-transportable configuration, marked on the silhouette views.

c. Shipping weight.

d. Loading cubage.

e. Overall height, width, and length.

f. Front and rear axle loads.

g. Tie down information.

h. Center of gravity.

2.2.3.3 Lubrication data plate. The aircraft staircase shall be provided with a permanently-marked, lubrication data plate constructed of a non-corroding metal, mounted on the exterior surface of the aircraft staircase and shall direct attention to all lubrication fittings and components requiring lubrication. The plate shall identify the type and grade of lubricant required for operational temperatures specified in 2.3.1.

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2.2.3.4 Decals. A warning decal shall be provided in the cab advising of the hazards of driving with stairs extended. This decal shall include speed limitations while the stairs are extended; in essence a tipping-over warning. Additionally, an overhead-clearance decal shall be provided in the cab in plain view of the driver or operator stating the minimum and maximum stair heights.

2.2.4 Environment, Safety, and Occupational Health (ESOH).

2.2.4.1 Safety. The aircraft staircase shall comply with all applicable requirements of Title 49 of the Code of Federal Regulations (49 CFR), part 393, Parts and Accessories Necessary for Safe Operation; 49 CFR, part 571, Federal Motor Vehicle Safety Standards; and Occupational Safety and Health Administration (OSHA) standards in effect at the time of manufacture. The aircraft staircase design shall not contain any mishap risk categories greater than medium as defined in MIL-STD-882, table III.

2.2.4.2 Hazardous material. The aircraft staircase design shall minimize and control hazards associated with the inclusion or use of hazardous or toxic materials and the generation of toxic or noxious gases. The aircraft staircase shall not generate or use class I or class II ozone depleting substances (ODS) during operation, maintenance, or disposal. Class I ODS and hazardous materials shall not be used in any system, component, or process. The aircraft staircase shall not contain or use either hexavalent chromium or cadmium without written approval by the procuring activity. Hazardous materials are defined in AFI 32-7086; class I and class II ODS are defined in 40 CFR, part 82, Protection of Stratospheric Ozone.

2.2.4.3 Foreign object damage (FOD). All loose metal parts, such as pins or connector covers, shall be securely attached to the aircraft staircase with wire ropes or chains. "Dog tag" style, beaded chains shall not be provided. Removable panels, if provided, shall be attached with captive fasteners. Tire valve-stem caps shall be made of plastic.

2.2.4.4 Noise levels. The aircraft staircase interior sound level shall be in accordance with 49 CFR, 393.94, Interior Noise Levels in Power Units.

2.2.4.5 Component protection. All space in which work is performed during operation, service, and maintenance shall be free of hazardous protrusions, sharp edges, or other features that may cause injury to personnel. All rotating and reciprocating parts, and all parts subject to high operational temperatures or subject to being electrically energized that are of such nature or so located as to be hazardous to personnel shall be guarded or insulated to eliminate the hazard. All wires, cables, tubes, and hoses shall be supported and protected to minimize chafing and abrasion, and shall be located so as to provide adequate clearance from moving parts and high operational temperatures. Grommets shall be provided wherever wires, cables, tubes, or hoses pass through bulkheads, partitions, or structural members.

2.2.4.6 Backing safety. The aircraft staircase shall be equipped with a flashing beacon and a siren that automatically engages when the vehicle is placed in reverse.

2.2.5 Human systems integration. The aircraft staircase shall be designed in accordance with MIL-STD-1472 for ease of operation, inspection, and maintenance, including the use of arctic mittens

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and mission-oriented protective posture (MOPP), level 4, chemical warfare gear. Chemical warfare gear is not required for preventive maintenance or major corrective maintenance.

2.2.6 Fastening devices. All screws, bolts, nuts, pins, and other fastening devices shall be properly designed, manufactured, and installed with adequate means of preventing loss of torque or adjustment. Cotter pins, lock washers, or nylon patches shall not be used for this purpose, except for the attachment of trim items or as provided in commercial components. Tapped threads on loaded fasteners shall have a minimum thread engagement in accordance with table I.

TABLE I. Minimum thread engagement.

Material	Minimum Thread Engagement
Steel	1.0 times the nominal fastener diameter
Aluminum, zinc, or plastic	2.0 times the nominal fastener diameter

2.2.7 Welds and welding. All welders shall be certified to weld in accordance with American Welding Society (AWS) D1.1 and AWS D1.2. The contractor shall make available to the Government certifications for all welders being utilized on the aircraft staircase. Welding procedures and all welding on the aircraft staircase shall be completed in accordance with AWS D1.1 and AWS D1.2. The surface parts to be welded shall be free from rust, scale, paint, grease, and other foreign matter. Welds shall be of sufficient size and shape to develop the full strength of the welded parts. Welds shall transmit stress without cracking or permanent distortion when the parts connected by the welds are subjected to test, proof, and service loadings.

2.3 Environmental conditions.

2.3.1 Operating temperature range. The aircraft staircase shall be capable of operating in ambient temperatures ranging from -40 to 125 °F.

2.3.2 Storage temperature range. The aircraft staircase shall be capable of being stored in ambient temperatures ranging from -60 to 160 °F.

2.3.3 Precipitation.

2.3.3.1 Rain. The aircraft staircase shall be capable of storage and operation during rainfall of 5-inches per hour for three consecutive hours and 10-inches per hour for 10 consecutive minutes with winds of up to 35 knots, and with 6-inches of rain per hour impinging on the aircraft staircase at angles from vertical to 45° for 30 consecutive minutes.

2.3.3.2 Snow. The aircraft staircase shall be capable of storage and operation during accretion of wet snow up to 2-inches per hour for at least 12 hours.

2.3.3.3 Ice. The aircraft staircase shall be capable of storage and operation with ice accretion up to 1.5-inches on exposed horizontal surfaces. An operator may use an ice scraper for five minutes during the start-up process.

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2.3.4 Solar radiation. The aircraft staircase shall not be adversely affected by full-time exposure to solar radiation, such as those conditions encountered in desert environments.

2.3.5 Fungus. All materials used in the aircraft staircase shall be fungus resistant or shall be suitably treated to resist fungus. Materials treated for fungus resistance shall retain their original electronic and physical properties, shall not present toxic hazards, and treatment shall last for the entire service life of the part. The aircraft staircase shall be suitable for operation and storage in conditions encountered in a tropical environment.

2.3.6 Salt fog. The aircraft staircase shall be capable of storage and operation in high temperature, high humidity, salt-laden, sea-coast environments without damage or deterioration of performance.

2.3.7 Sand and dust. The aircraft staircase shall be capable of storage and operation during exposure to wind-blown sand or dust without damage or deterioration of performance.

2.3.8 Wind stability. The aircraft staircase shall be stable, in accordance with SAE ARP1328, while the stairs are fully extended. The horizontal movement of the top platform shall not exceed 3 inches in either direction.

2.4 Transportability.

2.4.1 Surface transportability. The aircraft staircase shall be transportable via all modes of surface shipment (highway, rail, and water) in accordance with MIL-STD-1366, and shall be capable of withstanding the mechanical shock and vibration characteristics of highway, rail, and water transport, except that design for rail-impact testing (see MIL-STD-1366, 5.2.5) is not required.

2.4.2 Air transportability. The aircraft staircase shall be transportable without shoring on C-5 and C-17 aircraft. Design criteria can be found in MIL-STD-1791. In all air-transport configurations, the aircraft staircase shall be capable of being restrained and withstanding 2.0 G up and 4.5 G down accelerations without loss of serviceability, and shall be capable of being restrained and withstanding 3.0 G forward, 1.5 G aft, and 1.5 G lateral accelerations without loss of structural integrity. Axle weight shall not exceed 20,000 pounds. Tire pressure shall be less than 100 pounds per square inch (psi) and shall not be reduced for air transport. The aircraft staircase shall be equipped with pressure-relief devices or configured for air transport to prevent any part from becoming a projectile in the event of catastrophic loss of aircraft cabin pressure. The aircraft staircase shall drive on and off the aircraft, negotiating the required maximum ramp angles without shoring.

2.4.3 Equipment removal/reconfiguration. Preparation for air transport shall take no more than one hour for two persons using common, non-powered, hand tools (see 6.3.2). All equipment removed shall be stored on the aircraft staircase; caps and plugs shall permit moving and storage in transport configuration. The forces required for equipment removal/reconfiguration shall not exceed those allowed in MIL-STD-1472.

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2.4.4 Tie downs. The aircraft staircase shall be symmetrically restrained during air and ground transport. Tie down points shall be rated at a minimum of 25,000 pounds, marked for capacity, with a clear opening compatible with MIL-DTL-25959 and MIL-PRF-27260 tie down devices. Each end of each tie down device shall terminate at a tie down point and not pass through any other tie down point. There shall be no interference between tie down devices and the aircraft staircase. The tie down provisions shall be in accordance with MIL-STD-209, 4.1 through 4.12.

2.5 Maintainability. The aircraft staircase and all included systems shall be designed and constructed for maintainability in accordance with MIL-STD-1472, 5.9 through 5.9.18; forces shall not exceed those specified for both males and females, and shall require:

a. Where practical, parts and components shall be located or positioned for rapid and simple inspection, and recognition of excessive wear and potential failure.

b. Capability of performing maintenance with general-purpose tools and equipment normally available commercially. The use of special-purpose tools (see 6.3.6) and equipment shall be supplied by the contractor and subject to approval by the procuring activity.

2.6 Foolproof installation. Where improper installation of an item causes a malfunction, an asymmetric mounting system shall be provided, where practical, to ensure proper mounting of the item.

2.7 Performance.

2.7.1 Chassis. The aircraft staircase shall be a standard, current truck model offered by a commercial automotive manufacturer, modified as necessary to comply with the requirements herein.

2.7.2 Gross vehicle weight rating. The aircraft staircase shall have a gross vehicle weight rating (GVWR) of at least 15,000 pounds. The gross vehicle weight shall not exceed 95 percent of the GVWR.

2.8 Mobility.

2.8.1 Operating terrain.

2.8.1.1 Paved surfaces. The fully-equipped aircraft staircase, loaded to its GVWR, shall maintain a speed of 45 miles per hour (mph) on a dry, level, paved road for at least 3 miles.

2.8.1.2 Graded gravel. The fully-equipped aircraft staircase, loaded to its GVWR, shall traverse a graded gravel road (see 6.3.4) at an average speed of 20 mph.

2.8.2 Clearances. The fully-equipped aircraft staircase, loaded to its GVWR, shall have a ground clearance of at least 8 inches. The aircraft staircase shall be designed to enter (approach), negotiate (up or down), and leave (depart) a solid surface ramp having a slope of 20 degrees. Angle of approach, angle of departure, and ramp breakover angle shall be selected accordingly. The length

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of the incline shall not be less than 1.2 times the wheelbase of the aircraft staircase. Definitions of clearance terms shall be in accordance with SAE J1100.

2.8.3 Steering force. The force required to turn the wheels from lock to lock, with the aircraft staircase loaded to its GVWR and the engine at low idle speed, shall not exceed 15 pounds when measured at the steering wheel rim.

2.8.4 Parking brakes. The parking brakes shall hold the aircraft staircase motionless on a 20-degree longitudinal grade in both upgrade and downgrade directions with the transmission in the neutral position. The grade surface shall be dry, hard, and free from loose material. The force required to set the brakes under these conditions shall not exceed 55 pounds.

2.8.5 Service brakes. The service brakes shall be in accordance with the performance requirements of Federal Motor Vehicle Safety Standard (FMVSS), Number 105.

2.8.6 Transmission. The aircraft staircase shall be equipped with an automatic transmission which shall include a hydraulic torque converter and not less than five forward-gear ratios. The net torque capacity and the net power rating of the transmission shall exceed the output ratings of the engine.

2.8.7 Axles. The aircraft staircase shall be equipped with axles rated at least equal to the load imposed on each axle, measured at the ground, with the vehicle loaded to its GVWR.

2.8.8 Suspension. The aircraft staircase shall be equipped with a suspension system with components having a rated capacity at least equal to the load imposed on each member, measured at the ground, with the vehicle loaded to its GVWR.

2.8.9 Steering. The aircraft staircase shall be equipped with a power-assisted steering system, with manual steering available in the event of power-assist failure.

2.8.10 Brakes. The aircraft staircase shall be equipped with power-assisted, self-adjusting, hydraulic brakes on all wheels, with an antilock braking system (ABS), in accordance with FMVSS, Number 105.

2.8.11 Tires and wheels. The aircraft staircase shall be equipped with single front and rear tubeless, steel-belted radial tires with all-season type tread mounted on steel disc-wheel assemblies. Tire-and-wheel assemblies shall be identical at all wheel positions. Tires and wheels shall be in accordance with the Tire and Rim Association Year Book requirements for this application. Under no circumstance shall the vehicle maximum-governed speed exceed the speed rating of any of the furnished tires. A spare tire-and-wheel assembly shall not be provided.

2.9 Engine and related equipment. The aircraft staircase shall be equipped with a diesel engine approved by the Environmental Protection Agency (EPA). The waiver required to purchase the engines shall be coordinated through the USAF aircraft staircase contracting officer and engineer.

2.9.1 Export engine. The aircraft staircase shall be equipped with a diesel engine compatible with JP-8 and F-24 jet fuels (see 2.9.7.1). At a minimum, the engine shall meet a Tier 2-level exhaust

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and evaporative emissions standard (export engine equivalent) for on-highway engines. This represents the highest level of exhaust and evaporative emissions standards that will permit the use of high-sulfur diesel fuels. In accordance with 40 CFR, 85.1708 the Air Force possesses an EPA waiver to allow for deviation from EPA emissions guidelines. The waiver will be transferred from the Air Force to the engine manufacturer. The transfer process requires the engine manufacturer to supply a decal for the engine stating, “This engine has an exemption for national security under 40 CFR, 1068.225.”

2.9.2 Engine starting system.

2.9.2.1 Starter. The engine shall be equipped with a 12-volt DC electric starter.

2.9.2.2 Engine-starting aids. The engine shall start within 15 seconds of cranking in any ambient temperature within the required operating range of the aircraft staircase. Internal engine-starting aids, fluid-starting aids, and heat from the winterization system (see 2.18) may be used prior to and during the start period to facilitate engine starting under the conditions in table II.

TABLE II. Engine-starting aids.

Temperature Range	Starting Aids Permitted
40 through 125 °F	None
0 through 39 °F	Internal engine-starting aids and fluid-starting aids
-40 through -1 °F	Internal engine-starting aids, fluid-starting aids, and heat from the winterization system

2.9.3 Engine air intake system. The engine air intake system shall be in accordance with SAE ARP1247, 3.13.1.4.3. The inlet shall not draw air from directly beneath the aircraft staircase, and shall not be located near the cooling system air outlet nor the engine exhaust outlet. Joints shall be minimized between the air filter outlet and the actual engine air inlet, and shall be designed to ensure no leakage of unfiltered air into the engine. A differential pressure, air filter service indicator shall be provided.

2.9.4 Engine cooling system. The engine cooling system shall be in accordance with SAE ARP1247, 3.13.1.4.2. Silicone radiator and heater hoses, constant-torque clamps, a coolant filter, and a coolant recovery system shall be provided. Engine coolant shall be in accordance with A-A-52624, type I, and of adequate strength to provide protection to -40 °F. The engine out (top of radiator) coolant temperature shall not exceed 210 °F or the engine manufacturer’s recommendations at 125 °F. The coolant filter shall be rated for one year or 2,000 hours between replacements, with supplementary coolant additives (SCA) either included in the filter or added separately.

2.9.5 Engine lubrication system. The engine lubrication system shall be designed so that the aircraft staircase can be operated on a 20-degree longitudinal slope facing both up and down and on an 8-degree side slope in each direction (right and left side facing up the slope).

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2.9.5.1 Engine oil. The engine shall be compatible with Grade 15W40 of MIL-PRF-2104 from 0 to 125 °F. Oil pre-heat for operation below 0 °F is allowed. The engine shall be compatible with arctic engine oil from -40 to 60 °F in accordance with MIL-PRF-46167.

2.9.5.1.1 Engine oil operating temperature. The engine oil sump temperature shall not exceed 250 °F or the engine manufacturer's recommendation at 125 °F.

2.9.5.1.2 Engine oil consumption. The engine oil consumption shall not exceed 0.0035 pounds per brake horsepower-hour (lbs/bhp-hr) under any operating condition.

2.9.5.2 Engine oil filter. The engine oil filter shall be in accordance with SAE ARP1247, 3.13.1.4.4.

2.9.6 Exhaust system. The exhaust system shall be constructed of stainless steel. The muffler(s) shall be constructed of aluminized steel or stainless steel. Exhaust system outlet(s) shall be directed away from personnel accessing any control panel or equipment compartment and the engine air intake, and shall not be directed toward the ground.

2.9.7 Engine fuels and fuel system.

2.9.7.1 Engine primary fuels. The following shall be primary fuels (see 6.3.5):

a. JP-5, in accordance with MIL-DTL-5624, -40 to 125 °F ambient air temperature.

b. JP-8, in accordance with MIL-DTL-83133, -40 to 125 °F ambient air temperature.

c. F-24, in accordance with NATO AFLP-3747, -40 to 125 °F ambient air temperature.

2.9.7.2 Engine alternate fuels. The following shall be alternate fuels (see 6.3.1):

a. 1-D S15, 1-D S500, and 1-D S5000 in accordance with ASTM D975, below 60 °F ambient air temperature.

b. 2-D S15, 2-D S500, and 2-D S5000 in accordance with ASTM D975, above 32 °F ambient air temperature.

c. TS-1 in accordance with GOST 10227-86 with JP-8 additives, -40 to 125 °F ambient air temperature.

2.9.7.3 Engine emergency fuels. The following shall be emergency fuels (see 6.3.3): Jet A and Jet A1 in accordance with ASTM D1655-05, all ambient temperatures.

Note: Jet A and Jet A1 may contain corrosion inhibitor/lubricity improver and icing inhibitors which shall result in properties equal to JP-8.

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2.9.7.4 Engine fuel consumption. When operating on JP-8, the engine at full-rated load shall not exceed 0.8 lbs/bhp-hr brake specific fuel consumption.

2.9.7.5 Fuel system. The fuel system shall be in accordance with SAE ARP1247, 3.13.1.5.1 through 3.13.1.5.11, except as otherwise specified herein. The fuel system shall be constructed of materials which are compatible with the fuels listed in 3.9.6.1 and 3.9.6.2. Copper shall not be used in the fuel system.

2.9.7.5.1 Fuel filters. Primary and secondary fuel filters and a heated fuel/water separator shall be provided. The fuel/water separator shall include a water coalescer and a drain valve that is readily accessible by an operator or a mechanic. A combination fuel filter and fuel/water separator may be provided. Fuel filter elements shall be easily replaceable by a mechanic using nothing more than common hand tools (see 6.3.2) without loss of engine prime.

2.9.7.5.2 Fuel tank. The fuel tank shall be in accordance with SAE ARP1247, 3.13.1.5.5 through 3.13.1.5.9. The tank shall be designed so that the aircraft staircase can be operated on a 20-degree longitudinal slope facing up and down and on an 8-degree side slope in each direction (right and left side facing up the slope). The tank shall be provided with corrosion protection and baffles. A 0.25- to 0.375-inch nominal drain valve shall be provided for emptying fuel and sediment into a container underneath the aircraft staircase without removal of the tank or any other major component. The fuel tank shall have a fuel fill opening of not less than 3-inch inside diameter, and shall be designed to drain fuel spillage overboard for collection outside the aircraft staircase. The fuel cap shall be equipped with a retention device to prevent loss and FOD. The fuel fill opening, fuel cap, and fuel cap retention device shall be fabricated from non-sparking material.

2.9.8 Engine diagnostic and emergency shutdown systems.

2.9.8.1 Engine diagnostic system. If the engine is equipped with an electronic control module, a diagnostic system shall be provided with a means to indicate engine faults. If the aircraft staircase is equipped with a diagnostic or built-in-test system, the engine diagnostic system shall be integrated with it; if not, it shall be a stand-alone system.

2.9.8.2 Engine emergency shutdown system. The engine shall be equipped with an engine emergency-shutdown system consisting of a fuel-cutoff solenoid activated by the following conditions:

a. Low oil pressure, less than 10 psi or in accordance with the engine manufacturer's recommendations.

b. Coolant overheat, or cylinder head temperature overheat condition in accordance with the engine manufacturer's recommendations.

c. Engine over speed should the engine exceed 110 percent rated speed. This is required for engines over 100 bhp and optional for engines less than 100 bhp rated horsepower.

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2.9.9 Engine operator instruments. The following instruments, at the least, shall be provided to the engine operator:

a. Tachometer.

b. Speedometer/odometer.

c. Coolant-temperature gauge.

d. Low-coolant-level indicator.

e. Oil-pressure gauge.

f. Oil-temperature gauge.

g. Fuel-level gauge.

h. Hour meter.

i. Ammeter or voltmeter.

2.10 License-plate bracket. A lighted license-plate bracket with all applicable hardware shall be provided at the left rear of the vehicle. A second plate bracket and hardware shall be provided on the front bumper.

2.11 Cab. The aircraft staircase shall include an enclosed, air-conditioned and heated cab for the driver and one passenger. The staircase, when fully lowered and stowed, shall not obstruct visibility through the windshield. In addition to the standard, truck cab equipment (heater/defroster, lights, electric wipers and washer, rearview mirrors, etc.) the following items shall be provided:

a. A minimum 34-inch x 16-inch, glare-proof, waterproof observation window with an electric wiper to be fitted to the roof of the cab. The roof window shall be sufficiently large to permit the driver of the vehicle to view, from the normal driving position, the entire underside of the mating area of the aircraft being serviced from the normal driving position. In lieu of a window, a camera may be placed on the A-frame to provide a clear view of the selection positions of each model of aircraft. The camera shall provide a clear picture on a backlit LCD monitor in the vehicle cab. The monitor shall be a minimum 4-inch x 5-inch and be viewable in day and night operations.

b. Two heated side-view mirrors, approximately 17 inches long and 8 inches wide, with nominal inner-convex mirror.

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2.12 Staircase.

2.12.1 Aircraft interface. The aircraft staircase shall be designed to provide a safe means of access onto and off of the passenger entrance/exit of the aircraft. The top platform shall be adjustable to all heights above ground level within the range of 96 inches to 228 inches for interface with the following aircraft: B-707, B-727, B-737, B-747, B-757, B-767, B-777, A-300, A-320, A-340, and their military derivatives. The staircase shall allow the aircraft doors for all airframes listed to fully open and close with the staircase completely in place and fully engaged.

2.12.1.1 Aircraft mating. The ramp-adjustment provisions shall be so designed that the top platform and its sliding side panels will mate with aircraft loading doors at any height within the required operating range without forming a step or gap of more than 2.0 inches between the doorsill of the aircraft and the top platform. The platform side panels shall not strike the side of the aircraft serviced by this staircase, nor form a gap at the top of the panel between the side panel and aircraft of more than 3.0 inches when adjusted for the aircraft being serviced. The side panels shall lock in position with positive-type lock and hands-free operation.

2.12.1.2 Bumpers. A tubular, rubber bumper shall be securely fastened to the front of the top platform to protect the aircraft from possible damage when the staircase is firmly positioned against the aircraft. The outside diameter of the bumper shall be a minimum of 7.0 inches. Durable bumpers shall also be fastened on the front of the platform side panels to protect the aircraft from damage when the panels are extended and positioned against the aircraft.

2.12.2 Staircase operation. The aircraft staircase shall have the capability to safely drive short distances, up to 100 ft, with the stairs extended to each aircraft doorsill height requirement. The aircraft staircase shall be equipped with a governance that restricts speed to 5 mph when the stairs are extended. The control panel shall have an indicator light(s) to provide the operator real-time status of whether the lift is extended or retracted.

2.12.2.1 President of the United States operation. The aircraft staircase shall have the ability to be extended, adjusted to the operational height (182 inches), and angled at a prepositioned site. The vehicle shall be driven to the doorsill of the aircraft in this configuration. At no time will any portion of the asset (outriggers) drag the ground or inhibit operation while driving to the aircraft. All controls to perform this operation shall be automated at the cab's control panel, as necessary. Neither the operator nor mechanic shall be required to work outside the cab to prepare the truck for this operation.

2.12.3 Load capacity.

2.12.3.1 Concentrated load. Each step and the top platform of the staircase shall withstand a concentrated load of 500 pounds on any square foot with no permanent distortion.

2.12.3.2 Total static load. The fully-extended staircase, unsupported by aircraft contact, shall withstand a combined static load of 250 pounds on each step and 1,000 pounds on the platform. The total deflection of the staircase platform shall be less than 1.0 inch from the unloaded to fully-loaded condition.

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2.12.4 Stairs. The staircase shall be constructed of non-glare material. The steps and the platforms shall be drainable, aluminum diamond plate, or comparable non-slip design in accordance with SAE AIR6133. To prevent slipping, step surfaces shall have a coefficient of friction of at least 0.5 for level surfaces and 0.8 for sloped surfaces.

2.12.4.1 Riser and tread. Each step shall be at least 42 inches wide. Riser heights and tread depths shall be in accordance with SAE AIR6133, table 3, for fixed, riser-type steps. The step treads shall not deviate more than 3.0 degrees from the horizontal regardless of platform elevation. The step riser shall connect the front of each step with the rear of the next lower step at each platform height within the operating range. Step and step riser design shall ensure that usable tread has a minimum depth of 9.0 inches.

2.12.4.2 Platform. The top step shall be made in the form of a platform. The width of the platform shall be at least the width of the steps and the length shall be at least 58 inches. The platform shall incorporate a pivoting section to accommodate the gap caused by imperfect alignment between the platform and the various aircraft.

2.12.4.3 Bottom step. For other than loading operations, the bottom step shall be designed such that it can easily be positioned to provide the required ground clearance (see 2.8.2). The bottom step shall have a rise no greater than 7.0 inches for any configuration of the staircase and all airframe models.

2.12.5 Handrails. Staircase and platform handrails shall be capable of withstanding a force of 200 pounds in any direction. There shall be a clearance of at least 3.0 inches between the handrails and any other object, such as side panels (see 2.12.6). The handrails shall be stainless steel material and at least 1.5 inches in diameter.

2.12.5.1 Staircase handrails. Handrails shall be provided on each side of the staircase, extending from the base of the stairs to the aircraft end of the top platform. Handrail heights shall be in accordance with SAE AIR6133, table 3. The handrails shall maintain a constant height above the steps at all platform heights within the operating range of the staircase.

2.12.5.2 Platform handrails. The top platform shall have handrails that are at least 42 inches in height. The handrails on the platform shall allow opening and closing of the aircraft entrance doors with the staircase in place.

2.12.6 Side panels. Vertical siding shall be provided and shall extend upward a minimum of five-eighths the distance from the steps to the handrails, and five-eighths the distance from the bottom to the top of the platform side panels. It is preferred that the platform side panels be at least seven-eighths the vertical height of the panels. The side panels shall be capable of withstanding a 200-pound horizontal load at any point.

2.12.6.1 Platform side-panel rollers. If rollers are utilized on the top platform side panels, the rollers shall be configured to conform to the shape of the track on which they operate, and shall be of sufficient depth to prevent the rollers from coming off or out of the track. Platform side-panel

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rollers shall be fail-safe in that a redundant system shall be provided to ensure the panels remain in place under all operating conditions. Rollers shall be rustproof or similarly have sealed bearings.

2.12.7 Staircase removal provisions. The staircase shall be provided with means to enable it to be lifted from the chassis. If lifting eyes are used, the inside diameter of each lifting eye shall be a minimum of 2.0 inches. A minimum static safety factor of 3:1 shall be applicable for the lifting provisions, based on the weight of the staircase unit. A stencil, sticker, or other identifying plate shall indicate the staircase removal points as "Staircase Removal Points Only."

2.13 Staircase lift system. A hydraulic height-adjustment system shall be provided to elevate and lower the staircase through its entire operating range. The system shall be designed to operate the unloaded staircase from the lowest to the highest position in either direction in a maximum of two minutes. The system shall regulate rapid pressure drops when operating the lift system. Relief valves, preset to release at no lower than 125 percent of normal operating pressure of the circuit, shall be provided. The system shall show no leaks or breakage when the staircase is subjected to the loads specified herein. When so loaded, the platform shall not settle more than a 0.5 inch in a 4-hour period. The platform shall not sway more than 3 inches in any direction during setup when the platform is being raised or lowered. Operating controls shall be provided on the instrument panel within the operator's compartment, as well as on the platform to allow for raising and lowering the stairs. Positive removal stops shall be provided to prevent over extension of the upper ramp assembly.

2.13.1 Actuating pump. The lift system shall be powered by a hydraulic pump, which is directly driven by the chassis engine.

2.13.2 Auxiliary pump. An auxiliary pump powered by the chassis electrical system shall be provided to allow the staircase to be lowered and stowed, and the stabilizers raised and stored in the event of an actuating pump failure.

2.13.3 Lift lock. The aircraft staircase shall be provided with a positive mechanism to lock the staircase in any position attainable by the staircase lift. A release control shall be provided on the instrument panel to disengage the lift lock. If the lift lock is not automatically unlocked by actuation of the raising and lowering control, it shall be capable of withstanding the full load as imposed through the hydraulic system up to relief-valve pressures.

2.14 Hydraulic system design. The hydraulic system shall be in accordance with SAE ARP1247, 3.13.1.3, except as otherwise specified herein. O-ring face seal hydraulic fittings may be used in lieu of flared fittings (see SAE ARP1247, 3.13.1.3.12). All hydraulic system components, including the hydraulic tank, shall comply with all corrosion-resistance requirements specified herein. System design (seals, filters, etc.) should consider exposure to sand and dust particles as encountered in desert areas.

2.14.1 Hydraulic reservoir. The storage tank shall have a capacity of not less than 110 percent of the fluid required to operate the hydraulic system. The tank shall be equipped with a fluid level indicator, a drain plug, and an air vent. The hydraulic system and tank shall be fitted with hydraulic fluid in accordance with MIL-PRF-83282. A strainer shall be installed in the hydraulic suction

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line. A filter system shall be installed in the pump pressure line. The filter shall be so located as to provide access for element replacement. The degree of filtration shall be from 10 - 40 microns.

2.14.2 Hydraulic cylinders. With the possible exception of the main lift cylinder, standard commercial hydraulic cylinders shall be provided. Seals shall be installed to prevent the entrance of moisture into the cylinders, and all cylinders shall be protected with loaded lip seals or O-rings with backup rings.

2.15 Stabilizers/Outriggers. Hydraulic stabilizers/outriggers, spring lockouts, or other support devices may be used to establish rigidity; however, the use of such devices shall not change the platform height after initial positioning to the aircraft. Each retractable set (rear, center, and front) shall be independently operated from automated switches located on the cab's control panel. If stabilizers/outriggers are utilized, they shall be capable of being returned to the "travel" position in the event of power failure. They shall be capable of being positioned in 30 seconds and retracted in 30 seconds with power supplied. Stabilizers/outriggers shall remain either in the fully-retracted or fully-extended position when left for extended periods of time. They shall not exacerbate the effects of inclined surfaces, thereby, making setup more difficult and cumbersome. They shall provide for maximum stability even on inclined surfaces. They shall all make equal contact with the ground and thus provide allowance for uneven surfaces.

2.16 Electrical system. The aircraft staircase shall have a 12-volt electrical system in accordance with SAE ARP1247, 3.13.1.2, except where specified.

2.16.1 Alternator. The aircraft staircase shall be equipped with a single or dual alternator charging system rated at not-less-than 60-ampere, in accordance with SAE ARP1247. The alternator's output at engine idle speed shall provide for the continuous electrical load of the aircraft staircase. It shall be capable of restoring the energy expended during an engine start in less than 15 minutes of engine idle at -20 °F.

2.16.2 Batteries. The batteries shall be of the commercial maintenance-free type battery; addition of water shall not be required during normal service life. The battery cover and vent system shall be designed to prevent electrolyte loss during service, and to keep the top of the battery free from electrolyte. The battery box shall have a drain hole and a plug.

2.16.3 Battery compartment. The batteries shall be enclosed in a weatherproof box or compartment and be readily accessible. If the batteries are located in the engine compartment, a battery box shall not be required. The battery compartment shall not be near the fuel tank or any other fuel or hydraulic system component.

2.16.4 Wiring. Wiring, except that provided with the commercial chassis, shall be in accordance with the applicable portions of SAE ARP1247, 3.13.1.2. All wiring exposed to possible physical damage, such as wiring on a compartment floor, or wiring where tools or materials may cause the wire to be frayed or shorted shall be protected by the use of heavy-wall, thermoplastic conduit or steel conduit having waterproof connections. Grommets shall be used for all wiring passing through any hole or body framework.

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2.17 Lighting system.

2.17.1 Step and platform lighting. The staircase shall be equipped with sufficient lights to ensure the platforms and every step will be clearly visible at night to personnel using the staircase. The lights shall be recessed flush with the staircase panels, weatherproofed, and shall provide a minimum illumination of 10 foot-candles over the entire area to be traversed. The step and platform lights shall be controllable by switches in both the driver's cab and on the top platform. Ultra-bright LED lights shall be utilized.

2.17.2 Spotlights. Two adjustable, waterproof, commercial spotlights shall be provided of sufficient size and capacity to illuminate the underside of the platform, the bumper, the aircraft door, and adjacent area to facilitate positioning the staircase to the aircraft. The spotlights shall be controllable by switches in both the cab's control panel and on the top of the platform.

2.18 Winterization system. When specified (see 6.2), a winterization system consisting of heaters for engine coolant, oil pan, transmission, and hydraulic system, as well as battery warmers, shall be provided for operation in temperatures down to -40 °F. The optional winterization system shall be designed to operate from an external 110-volt AC power source, and shall incorporate high-temperature shutoff switches to prevent overheating of any fluid or component.

2.19 Workmanship. The aircraft staircase, including all parts and accessories, shall be constructed and finished in a thoroughly workmanlike manner in accordance with SAE ARP1247, 3.13.7. Workmanship objectives shall include freedom from blemishes, defects, burrs, and sharp corners and edges; accuracy of dimensions, surface finish, and radii of fillets; thoroughness of soldering, welding, brazing, painting, wiring, and riveting; marking of parts and assemblies; alignment of parts and tightness of assembly fasteners; etc.

2.20 Bolted connections. Bolt holes shall be accurately punched or drilled, and they shall be deburred. Threaded fasteners shall be tight and shall not work loose during testing or service use.

2.21 Riveted connections. Rivet holes shall be accurately punched or drilled, and shall be deburred. Rivets shall be driven with pressure tools and shall completely fill the holes. Rivet heads shall be full, neatly made, concentric with the rivet holes, and in full contact with the surface of the component.

2.22 Gear and lever assemblies. Gear and lever assemblies shall be properly aligned and meshed, and shall be operable without interference, tight spots, loose spots, or other irregularities. Where required for accurate adjustment, gear assemblies shall be free of excessive backlash.

2.23 Cleaning. The aircraft staircase shall be thoroughly cleaned. Loose, spattered, or excess solder; welding slag; stray bolts, nuts, and washers; rust; metal particles; pipe compound; and other foreign matter shall be removed during and after final assembly.

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3. REGULATORY REQUIREMENTS.

3.1 Recycled, recovered, or environmentally preferable materials. Recycled, recovered, or environmentally preferable materials should be used to the maximum extent possible provided that the material meets or exceeds the operational and maintenance requirements, and promotes economically advantageous life-cycle costs. The offeror/contractor is encouraged to use recovered materials or biobased products to the maximum extent practicable, in accordance with Federal Acquisition Regulation (FAR), part 23.403. However, used, rebuilt, or refurbished items shall not be provided.

3.2 Green Procurement Program. Green Procurement Program (GPP) is a mandatory federal acquisition program that focuses on the purchase and use of environmentally preferable products and services. GPP requirements apply to all acquisitions using appropriated funds, including services and new requirements. FAR 23.404(b) applies and states the GPP requires 100 percent of EPA-designated product purchase that are included in the Comprehensive Procurement Guidelines list that contains recovered materials, unless the item cannot be acquired: a) competitively within a reasonable timeframe; b) meeting reasonable performance standards; or c) at a reasonable price. The prime contractor is responsible for ensuring that all subcontractors comply with this requirement.

4. PRODUCT CONFORMANCE. The products provided shall meet the salient characteristics of this CID; conform to the producer's own drawings, specifications, standards, and quality assurance practices; and be the same product offered for sale in the commercial marketplace, modified as necessary to comply with the requirements herein. The Government reserves the right to require proof of such conformance.

4.1 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First production inspection.

b. Conformance inspection.

4.1.1 First production inspection. The first production aircraft staircase shall be subjected to the examinations, demonstrations, tests, and analyses described in 4.4. The contractor shall provide or arrange for all test equipment and facilities. Except as otherwise specified, all testing in which engine is operated shall be performed using JP-8 turbine fuel.

4.1.2 Conformance inspection. Each production aircraft staircase shall be subjected to the examinations described in 4.4.

4.2 Commercial item requirement. The vehicle furnished shall comply with the "commercial item" definition of FAR 2.101 as of the date of award. The Government reserves the right to require the offeror/contractor to prove that their product complies with the referenced commerciality requirements and each salient characteristic of this CID.

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4.3 Inspection requirements.

4.3.1 General inspection requirements. Apparatus used in conjunction with the inspections specified herein shall be laboratory precision-type, calibrated at proper intervals to ensure laboratory accuracy.

4.3.2 Test facilities. All verification methods, test facilities, analyses, and certifications shall be approved by the Government Technical Point of Contact prior to the start of testing. The contractor is to receive DoD test rates at a Major Range Test Facility Base.

4.3.3 Test fuel. The preproduction test fuel requirement shall be Grade JP-8 or F-24 (see 2.9.7.1) turbine fuel. An exception may be taken for truck engine fuel during cold-chamber tests, as applicable. All filter elements shall be certified for use with JP-8 or F-24.

4.3.4 Data. During all testing specified herein, at least the following data, unless not applicable, shall be recorded at intervals not to exceed 30 minutes. Additional data and/or shorter intervals shall be provided as appropriate for any specific test.

a. Date.

b. Time started.

c. Time finished.

d. Ambient temperature.

e. Engine speed.

f. Engine oil pressure.

g. Engine coolant temperature.

h. Engine hour-meter readings at the start and finish of each test.

4.3.5 Test rejection criteria. Throughout all tests specified herein, the aircraft staircase shall be closely observed for the following conditions, which shall be cause for rejection:

a. Failure to conform to design or performance requirements specified herein or in the contractor's technical proposal.

b. Any spillage or leakage of any liquid, including fuel, coolant, lubricant, or hydraulic fluid under any condition, except as allowed herein.

c. Structural failure of any component, including permanent deformation or evidence of impending failure.

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d. Evidence of excessive wear.

e. Interference between the truck components or between the truck, the ground, and all required obstacles, with the exception of normal contact by the tires.

f. Misalignment of components.

g. Evidence of undesirable roadability characteristics, including instability in handling during cornering, braking, and while traversing all required terrain.

h. Conditions that present a safety hazard to personnel during operation, servicing, or maintenance.

i. Overheating of the engine or any other aircraft staircase component.

j. Evidence of corrosion or deterioration.

k. Failure of the hydraulic lift system.

4.3.6 Rejection and retest. Provided the aircraft staircase fails conformance testing or inspection, the aircraft staircase will be considered rejected until the cause is determined and corrective action is taken. Only then shall retesting be accomplished.

4.4 Test methods.

4.4.1 Examination of product. Each aircraft staircase shall be examined to determine compliance with the requirements herein, as well as SAE AIR6133 and SAE ARP1247. A contractor-generated, Government-approved checklist shall be used to identify each requirement not verified by an analysis, certification, demonstration, or test, and shall be used to document the examination results. Particular attention shall be given to materials, workmanship, dimensions, surface finishes, protective coatings and sealants and their application, welding, fastening, and markings. Proper operation of each aircraft staircase function shall be verified. Certifications and analyses shall be provided in accordance with table III. Each production aircraft staircase shall be inspected to a Government-approved reduced version of the checklist.

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TABLE III. Certifications and analyses.

Paragraph	Required Certifications and Analyses
2.4.1 Surface transportability.	Contractor surface transportability analysis (see 4.4.6.1) and certification that the aircraft staircase is transportable via all modes of surface shipment (highway, rail, and water) in accordance with MIL-STD-1366, and shall be capable of withstanding the mechanical shock and vibration characteristics of highway, rail, and water transport.
2.4.4 Tie downs.	Contractor tie down provision analysis (see 4.4.6.2.3).
2.4.2 Air transportability.	Contractor air transportability analysis (see 4.4.6.2.1) with ATTLA certification.
2.9 Engine and related equipment.	Engine manufacturer certification that the engine is in accordance with all applicable requirements, including exhaust emissions standards and fuels. Fuel certification shall verify the engine's compatibility with high-sulfur fuels (3000 parts per million); proof of acceptable sulfur range in the fuel. Engine manufacturer application approval for the engine and its installation, including cooling system.

4.4.2 Noise-level test. The first production aircraft staircase shall be tested to demonstrate compliance with 2.2.4.4.

4.4.3 Environmental testing.

4.4.3.1 High-temperature storage and operation test. A first production aircraft staircase shall be tested in accordance with MIL-STD-810, Method 501.7, procedures I and II, to demonstrate compliance with the high-temperature storage and operating requirements of 2.3.1 and 2.3.2. Test duration shall be one 24-hour cycle for each procedure beginning no less than two hours after test-item temperature stabilization.

4.4.3.2 Low-temperature storage and operation test. A first production aircraft staircase shall be tested in accordance with MIL-STD-810, Method 502.7, procedures I and II, to demonstrate compliance with the low-temperature storage and operating requirements of 2.3.1 and 2.3.2, as well as the engine starting requirement of 2.9.2 and the alternator charging requirement of 2.16.1. Test duration shall be one 24-hour cycle for each procedure beginning no less than two hours after test-item temperature stabilization. The winterization system shall be utilized during low-temperature operation testing to demonstrate compliance with 2.18.

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4.4.3.3 Precipitation.

4.4.3.3.1 Rain test. A first production aircraft staircase shall be tested in accordance with MIL-STD-810, Method 506.6, procedure I, to demonstrate compliance with 2.3.3.1 and 2.2.1.5.

4.4.5 Weight and center of gravity test. The weight, center of gravity, and axle weights of a first production aircraft staircase shall be measured to demonstrate compliance with the weight requirement of 2.7.2 and the axle-weight requirement of 2.8.7.

4.4.6 Transportability verification.

4.4.6.1 Surface-transportability verification. An engineering analysis shall be performed to demonstrate compliance with 2.4.1. The engineering analysis shall utilize the data for road transportation in accordance with MIL-STD-810, Method 514.8, table 514.8F II.

4.4.6.2 Air-transportability verification.

4.4.6.2.1 Air-transportability analysis. An engineering analysis shall be performed to demonstrate compliance with the air-transportability requirements of 2.4.2 and 2.4.3. The analysis shall include the tie downs and all major components and their ability to withstand the accelerations specified in 2.4.2. The evaluation shall also include a dimensional analysis for the aircraft staircase while traversing the ramp and while loaded aboard C-5 and C-17 aircraft.

4.4.6.2.2 Equipment removal and reconfiguration demonstration. A first production aircraft staircase shall be configured for transport on C-5 and C-17 aircraft, then reconfigured for operation to demonstrate compliance with 2.4.3. It shall be demonstrated that the forces required do not exceed those allowed in MIL-STD-1472.

4.4.6.2.3 Tie down provision analysis. An engineering analysis shall be performed to demonstrate compliance with the tie down provision requirements of 2.4.4.

4.4.6.2.4 Tie down provision test. A first production aircraft staircase shall be tested to demonstrate compliance with the tie down provision requirements of 2.4.4.

4.4.7 Maintainability demonstration. All recommended preventive maintenance tasks shall be performed, and the task times shall be recorded. The recommended frequencies of the preventive maintenance tasks and the times recorded to accomplish the tasks shall be used to develop an expected value of preventive maintenance time per measure of use, such as calendar time or hours of operation. It shall be demonstrated that the forces required do not exceed those allowed in MIL-STD-1472. All preventive maintenance tasks recommended to be performed daily and at the routine PMI shall also be performed by personnel wearing arctic mittens.

4.4.8 Mobility tests.

4.4.8.1 Paved road. A first production aircraft staircase, loaded to its GVWR, shall be driven at its maximum speed of 45 mph for a minimum of 3 miles on a dry, level, paved surface to demonstrate compliance with 2.8.1.1.

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4.4.8.2 Gravel roads. A first production aircraft staircase, loaded to its GVWR, shall be driven over gravel roads for a distance of at least 2 miles at an average speed of at least 20 mph to demonstrate compliance with 2.8.1.2.

4.4.8.3 Clearances. A first production aircraft staircase, loaded to its GVWR, shall be parked on a flat, level surface and its ground clearance, angle of approach, angle of departure, and ramp breakover angle shall be measured to demonstrate compliance with 2.8.2.

4.4.8.4 Steering force. A first production aircraft staircase, loaded to its GVWR and with the engine at low idle speed, shall be tested to demonstrate compliance with 2.8.3.

4.4.8.5 Parking brakes. A first production aircraft staircase, loaded to its GVWR, shall be tested in both directions on a dry, hard grade, free from loose material with the transmission in the neutral position to demonstrate compliance with 2.8.4.

4.4.8.6 Service brakes. A first production aircraft staircase, loaded to its GVWR, shall be tested on a level, dry, hard surface, free from loose material to demonstrate compliance with the stopping distance requirement of 2.8.5. At least four stops in opposite directions shall be made; two stops in each direction. The brakes may be allowed to cool to ambient temperature between stops.

4.4.9 Stability verification.

4.4.9.1 Stability analysis. An engineering analysis shall be performed to demonstrate compliance with SAE ARP1328 to demonstrate compliance with 2.3.8.

4.4.9.2 Stability test. The first production aircraft staircase, with the stairs fully extended, shall be tested in accordance with SAE ARP1328 to demonstrate compliance with 2.3.8.

4.4.10 Staircase operation test. The first production aircraft staircase shall be tested to demonstrate compliance with 2.12 through 2.18.

4.4.11 Load-capacity tests.

4.4.11.1 Staircase concentrated load test. The first production aircraft staircase shall be tested to demonstrate compliance with 2.12.3.1. Each load shall be applied for 15 minutes. The surface distortion shall be measured to the nearest hundredth decimal.

4.4.11.2 Staircase static-load test. The first production aircraft staircase shall be tested to demonstrate compliance with 2.12.3.2. The loads shall be applied for 24 hours. The total deflection shall be measured to the nearest tenth of a decimal.

4.4.11.3 Handrail load test. The first production aircraft staircase shall be tested to demonstrate compliance with all handrails under 2.12.5. The loads shall be applied for one hour.

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4.4.11.4 Side-panel load test. The first production aircraft staircase shall be tested to demonstrate compliance with 2.12.6. The loads shall be applied for one hour.

4.4.12 Staircase removal provision analysis. An engineering analysis shall be performed to demonstrate compliance with 2.12.7.

4.4.13 Welds and welding. Welds shall be inspected to ensure proper penetration and verified to be in strict accordance with the visual inspection criteria set forth in the AWS codes. The inspection shall be performed by an AWS Certified Weld Inspector (CWI) during and after the construction of all aircraft staircases. Documentation of all inspections for the first production aircraft staircase shall be provided to the Government. The Government reserves the right to require verification inspection at any time during construction in accordance with AWS D1.1, 6.1.2.2; AWS D1.2, 5.1.1; and non-specified, non-destructive test (NDT), other than visual, in accordance with AWS D1.1, 6.6.5. All welding documentation for subsequent production units shall be made available to the Government at any time during the life of the contract pursuant to AWS D1.1 and D1.2.

5. PACKAGING.

5.1 Preservation and packaging. Preservation, packing, and marking shall be as specified in the contract or order.

6. NOTES.

6.1 Source of documents.

6.1.1 Government documents. Copies of specifications, standards, and handbooks are available from DLA Document Services, 700 Robbins Avenue, Building 4/D, Philadelphia, PA 19111-5094, or online at https://quicksearch.dla.mil/.

6.1.2 Federal Acquisition Regulations (FAR). FARs may be obtained from the Superintendent of Documents, Government Printing Office, Washington, DC 20402, or online at https://www.acquisition.gov/?q=browsefar.

6.1.3 American Welding Society (AWS) documents. Copies of AWS documents may be obtained from American Welding Society, 8669 NW 36 Street, # 130, Miami, FL 33166-6672, or online at http://www.aws.org.

6.1.4 Society of Automotive Engineers (SAE) standards. Copies of SAE standards may be obtained from the Society of Automotive Engineers, 400 Commonwealth Drive, Warrendale, PA 15096, or online at http://www.sae.org/.

6.1.5 Code of Federal Regulations (CFR). CFRs may be obtained from the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402, or online at http://www.ecfr.gov/cgi-bin/ECFR?page=browse.

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6.1.6 Tire and Rim Association (TRA) standards. Copies of TRA standards may be obtained from the Tire and Rim Association, Inc., 175 Montrose West Avenue, Suite 150, Copley, OH 44321, or online at http://www.us-tra.org/publications.html.

6.1.7 Occupational Safety and Health Association (OSHA). Copies of OSHA documents may be obtained from Occupational Safety and Health Association, 200 Constitution Ave., NW, Washington, D.C. 20210, or online at http://www.osha.gov/.

6.1.8 Gosudarstvennye Standarty State standard (GOST). Copies of GOST 10227-86 are available from GOST Interstandard, 117049 Moscow, B-29 Russia, Attn: Mr. Nickolay Usov, nusov@gost.ru, or in English from www.ihs.com.

6.2 Ordering data. Acquisition documents must specify the following:

a. Title, number, and date of this purchase description.

b. When tropical rustproofing is required (see 2.2.1.4).

c. When winterization system is required (see 2.18).

6.3 Definitions.

6.3.1. Alternate fuel. A fuel that can be used in place of the primary fuel with a possible loss of efficiency; engine adjustments may be necessary. The use of an alternate fuel may result in a change of maintenance or overhaul cost.

6.3.2 Common hand tool. A non-powered tool that is likely to be found in a typical mechanic's toolbox. Common hand tools include open-end, box-end, combination, socket (both 6- and 12-point in both standard and deep-well), and hex-key wrenches, in SAE sizes up to and including 1 inch and metric sizes up to and including 25 mm; ratchet handles, extensions, and swivels; slotted and Phillips-head screwdrivers; regular and snap-ring pliers; and a ball-peen hammer.

6.3.3 Emergency fuel. A fuel that can be used when the primary and alternate fuels are not available. This fuel is not for use on a continuing basis, but is to be used only when the primary, alternate fuel, and equipment fuels are unobtainable and operation is mission essential. Engine life could be severely impacted.

6.3.4 Gravel road. A gravel road is a level or rolling gravel trail.

6.3.5 Primary fuel. The fuel or fuels used during tests to demonstrate system performance (contract compliance). Substitutions are presented as alternate and emergency fuels.

6.3.6 Special tool. A tool that is not commercially and readily available from a source other than the aircraft staircase contractor.

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6.4 Key Words.

Passenger
Type I
Work Vehicle
Airstair

MILITARY INTERESTS

Custodian:
Air Force – 184

Preparing activity:
Air Force – 184

(Project 1730-2019-002)

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using the ASSIST Online database at https://assist.dla.mil.

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